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MILITARY SPECIFICATION

COMPRESSOR UNIT, ROTARY: AIR, WHEELBARROW-FRAME-MOUNTED, TWO PNEUMATIC TIRES, GASOLINE-ENGINE-DRIVEN, 60-CFM, 6.5-PSIG

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a gasoline-engine-driven, rotary-type, air compressor rated at 60 standard cubic feet per minute (scfm) at 6.5 pounds per square inch gage (psig).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
WW-C-633	- Couplings, Hose, Pneumatic, Universal Type.
ZZ-H-500	- Hose, Rubber, and Hose Assemblies, Rubber; Pneumatic (Yarn or Fabric Reinforced).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173	- Varnish, Moisture-and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fittings, Lubrication.
MIL-C-3600	- Compressor, Rotary, Power-Driven; and Compressor, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1227	- Engine, Gasoline: Air-Cooled, 3 BHP, 4-Cycle, Military Design, Model 2A016, Installation Procedures.
MIL-STD-1474	- Noise Limits for Army Materiel.

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HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives, Preservatives and Specialty Products for use in Ground Equipment Systems.

2.1.2 Other Government document, drawing, and publication. The following other Government document, drawing, and publication form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 204 - Noise Emission Standards for Construction Equipment.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

DRAWING

ME

13214E6090 - Cap, Valve.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code for Displacement Compressors, Vacuum Pumps, and Blowers, PTC-9.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

ANSI/AWS D1.1 - Structural Welding Code, Steel.
ANSI/AWS D1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The compressor unit (hereinafter called "compressor") shall be portable and shall consist of a gasoline engine, coupling, compressor pump, fuel storage tank, air discharge manifold, air-relief valve, mounting frame, air-delivery hoses, and all other components and accessories necessary to make it complete and self-contained. The compresor pump shall be directly driven through a coupling to the engine and shall produce a minimum of 60 scfm at 6.5 psig at an engine speed of 3,600 revolutions per minute (rpm). The dry weight of the complete unit (excluding hoses) shall not exceed 160 pounds, based on engine weight conforming to MIL-STD-1227 (see 6.8).

3.2 First article. The contractor shall furnish one or more compressors for examination and testing within the time frame specified, to prove prior to starting production, that his production methods and choice of design detail will produce compressors that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.4). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests, as specified (see 6.2).

3.3 Initial production. When specified, the contractor shall furnish to the Government one or more compressors for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all the provisions of this specification (see 6.7).

3.4.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the air compressor may be newly fabricated from recovered materials, provided the air compressor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the air compressor.

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3.4.2 Material deterioration and control. The air compressor shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the air compressor may be exposed.

3.4.3 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.4.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.5 Environmental conditions.

3.5.1 Operation. The compressor shall produce not less than rated capacity and pressure at any temperature from -25° F to +120° F (see 3.8.1).

3.5.2 Precipitation. With rain falling at a rate of 3 inches per hour, the compressor shall perform as specified herein.

3.6 Safety. All parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All moving parts which are hazardous to personnel shall be enclosed or guarded. Exhaust or discharges from the compressor shall be directed so that they do not endanger personnel. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Protective devices shall not impair the operating functions. Warnings shall be mounted on or near components containing hidden hazards. Engine starting pulleys shall not be guarded.

3.6.1 Noise limits.

3.6.1.1 Sound limits. The noise level when delivering rated capacity shall not exceed 85 dB(A) at 15 feet when tested in accordance with 4.6.2.14. The distance from the control panel at which the noise level is 85 dB(A) shall be determined. That distance shall be shown on warning plates mounted on both sides of the compressor, stating "Warning. Ear Protection Required within _____ feet". The warning shall be legible from the distance where the level is met.

3.6.1.2 Personnel noise. If noise level exceeds 85 dB(A) at the control station when tested in accordance with 4.6.2.14, a feasibility study in accordance with MIL-STD-1474, 5.1.1.2, and cost estimate for reducing noise level to 85 dB(A) shall be provided.

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3.6.1.3 Exterior noise. The compressor shall meet the requirements of EPA 40 CFR 204, when tested in accordance with 4.6.2.14.1.

3.7 Maintainability. The compressor shall operate as specified herein without repair or maintenance other than the contractor's recommended normal schedule adjustments and servicing, as established by a maintenance schedule prepared and submitted by the contractor prior to test. All major assemblies and installed attachments shall be accessible for maintenance repair and replacement without removal of other major assemblies and installed attachments not normally removed. Cover plates which must be removed for component adjustment shall be equipped with quick-disconnect fastenings. All fasteners shall be of corrosion-resistant material or shall be treated to be corrosion resistant. All screw threads shall be in accordance with FED-STD-H28. Maximum use shall be made of interchangeable hardware and fastening devices. Means for drainage of lubricants, fuel, and condensate traps shall be in an accessible location and shall drain without splashing on any component or on operating personnel. The drain outlets shall be so located to facilitate complete drainage. Each maintenance, assembly or disassembly operation performed as a result of testing shall be accomplished by not more than two men using common tools and special tools furnished with the compressor.

3.8 Performance.

3.8.1 Rated capacity. The rated capacity of the compressor at an engine speed of 3,600 rpm shall be not less than 60 scfm at 6.5 psig measured at the manifold with air intake filter installed. The compressor shall operate for 500 hours without overhaul and shall produce rated capacity at 3,600 rpm when tested in accordance with 4.6.2.11. The maximum temperature rise between the compressor intake and discharge manifold shall not exceed 100° F.

3.8.2 Shock resistance. The compressor shall withstand the shocks imposed when dropped from a height of not less than 18 inches onto a level concrete surface without any misalignment, permanent deformation, or damage.

3.8.3 Tilted operation. The compressor shall deliver rated capacity when operated at rated speed and pressure, tilted 15 degrees from the horizontal in any plane.

3.9 Pneumatic system.

3.9.1 Compressor pump. The compressor manufacturer shall provide data to demonstrate that the compressor pump requires no more than 3.5 horsepower (hp) at an engine speed of 3,600 revolutions per minute (rpm) to produce the output capacity specified in 3.8.1. The data shall be made available for review by the contracting officer or the contracting officer's representative.

3.9.2 Compressor air-intake cleaner. A dry-type air-intake cleaner shall be provided. The air cleaner shall be securely attached and braced where required to prevent damage from vibration and shall be located so as to minimize foreign

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matter being drawn into the filter. The air cleaner shall be provided with a rain shield which shall prevent water from entering the air cleaner under conditions specified in 3.5.2.

3.9.3 Air-discharge manifold. An air-discharge manifold shall be provided to receive the compressed air directly from the compressor pump. The manifold shall have an integral air-relief valve and two discharge ports fitted with hose couplings conforming to WW-C-633, type III, size 1 inch. Both discharge port couplings shall be furnished with caps (dead-end couplings) conforming to WW-C-633, type V. The caps shall be chained to the manifold to prevent loss. The manifold shall permit passage of air with a minimum of friction and turbulence.

3.9.4 Air relief valve. The air-relief valve shall discharge 100 percent of the air at a manifold pressure not greater than 7.5 psig, and shall discharge not more than 0.25 cfm of free air at a manifold pressure of 5.0 psig.

3.9.5 Air-delivery hose. Two 25-foot lengths of hose conforming to ZZ-H-500, having an inside diameter of 1-1/4 inches, shall be furnished. Each hose shall have a steel adapter as shown in figure 1 at each end clamped to the hose by a ferrule of brass or corrosion-resistant steel, or by a corrosion-resistant steel hose clamp. The hose shall cover all of the adapter barbs, but no part of the hose, the ferrule, or clamp shall be within 1/8-inch of the threaded portion of the adapter. A hose coupling conforming to WW-C-633, type II, size 1 inch, shall be installed on one end of each hose.

3.9.6 Nozzles and inflating valves. Two aluminum nozzles shall be furnished with each compressor and shall be as shown in figure 2. Two inflating valves of the normally-closed, push-to-inflate type shall be furnished with each compressor, which shall assemble with the adapters shown in figure 1. The interior of the valves shall be designed to cause a minimum restriction of airflow. Each valve shall have a nozzle with a neoprene seal that shall be airtight at 7.5 psig when held against the open end of the valve cap shown on drawing 13214E6090. The gripping surface of the valves shall be a minimum of 2-1/4 inches in length and shall be completely or partially ribbed or knurled to prevent slipping. Ferrous metal, where used shall be cadmium plated (see 6.6).

3.10 Engine. A 3-horsepower military design engine, Model 2A016, will be provided by the Government for each compressor (see 6.3). The contractor shall install the engine in accordance with MIL-STD-1227.

3.10.1 Fuel tank. A fuel tank shall be provided and shall have a fuel capacity for not less than 4 hours of operation at compressor rated load. The filler neck opening shall be a minimum internal diameter of 1-1/2 inches, with a removable strainer, and a chain-secured filler cap. The filler cap shall be a screw cap that shall prevent fuel from leaking, when the compressor is tilted 15 degrees from the horizontal in any plane and when it is tilted for wheeled movement. The tank shall be provided with an air vent and drain. The fuel tank outlet shall be not less than 1/4 inch above the bottom of the fuel tank. The tank and all lines shall be located or positioned to eliminate fire hazard

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caused by spilling, overflow, or drainage of fuel. Construction and mounting shall be such that vibration, excessive tightening of the mounting devices, or contact with other components of the engine or compressor will not cause leaks to develop.

3.11 Toolbox. The compressor shall be provided with a toolbox made of steel 0.047 inch or aluminum 0.064 inch minimum nominal thickness. The toolbox shall have a hinged lid and a drawbolt or latch of a type that will keep the lid tightly closed when the toolbox is subjected to vibration. The toolbox shall be securely fastened to the compressor in a protected position and shall be provided with drain holes in the bottom corners. The toolbox shall be of sufficient size to accommodate the accessories (except hose) and tools furnished with the compressor.

3.12 Body frame structure. All components of the compressor, except wheels and air delivery hose, shall be mounted within a wheelbarrow-type body frame structure. The body frame structure shall not exceed 30 inches wide by 46 inches long by 36 inches high. Bends shall be continuous, symmetrical, and free of buckling. The structure shall be braced and provisions shall be made for shockproof mountings for compressor pump and engine. The structure shall be provided with handles located to permit lifting and carrying by two men or wheeling by one man. Means shall be provided for securing and carrying the discharge hoses on the compressor. The structure shall be mounted on two wheels having 4.10-4, 2 ply rating, industrial tires. Tires provided under this specification shall be not more than 12 months old on the date of shipment to the Government by the compressor manufacturer. The wheel assemblies shall be removable and mounted on a steel axle. Each wheel assembly shall be provided with two antifriction bearings conforming to FF-B-171, and provisions shall be made for securing the wheels in such a manner as to prevent the inner races of the bearings from rotating on the axles. The location of the wheels shall provide maximum ground clearance when the compressor is being moved. The longitudinal axis of the engine and compressor pump shall be in the horiztonal plane when the compressor is resting on a level surface.

3.13 Electromagnetic interference. The electromagnetic interference emission characteristics of the compressor shall conform to MIL-STD-461, class III C.

3.14 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.14.1 Lubricants. The compressor shall operate as specified herein when lubricated with military lubricants (see 6.5).

3.14.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

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3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.14.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and shall contain a drain plug. The drain outlet shall be located so that removal of tbe plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accessible for drainage to a container when the compressor is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet.

3.14.5 Grease lubrication. All grease lubrication, excluding lubrication of sealed bearings, shall be with grease as follows:

a. MIL-G-23827 - For instruments and bearings where operating temperatures do not exceed 250° F.
b. MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F.

The compressor shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.14.6 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.15 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II as applicable, 7 percent salicylanilide (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from

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from all foreign matter which would interfere with the adherence or function of the varnish.

3.15.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that composition II material complies with Rule 442 - Usage of Solvents.

3.16 Identification marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.17 Instruction plates. The compressor shall be equipped with instruction plates including diagrams, warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the compressor. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous and protected location.

3.18 Treatment and painting. The portions of the compressor normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.19 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished by the contractor (see 6.2).

3.20 Items to be furnished with compressors. No shipment of compressors shall be made unless repair parts, maintenance tools, and accessories are included with the shipment or unless approval for shipment of the compressors without such items has been received from the contracting officer.

3.21 Workmanship.

3.21.1 Castings and forgings. All parts, components, and assemblies of the compressor which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Re-work shall be limited to procedures which do not reduce strength or affect function.

3.21.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid

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overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.21.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to service loading.

3.21.3.1 Welders. Before assigning any welder to welding work covered by this specification, the contractor shall obtain certification that the welder and automatic welder equipment has passed qualification tests as prescribed by either of the following listed codes for the materials required to be welded and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1 - Structural Welding Code, Steel.
ANSI/AWS D1.2 - Structural Welding Code, Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.21.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.21.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 First article examination. The compressor shall be examined for the defects listed in table I. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The compressor shall be subjected to the tests marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production compressors will be selected at random by the Government from the compressors being produced by production tooling and will be examined for defects listed in table I and subjected to the tests marked "X" in column 3 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production compressor shall not exclude the remaining compressors from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all compressors accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each compressor shall be examined in accordance with 4.6.1. Presence of one or more defects shall be cause for rejection

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4.5.2 Tests. Each compressor shall be subjected to the test marked "X" in column 4 of table II. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. Examination shall be in accordance with table I.

TABLE I. Examination schedule.

Defect no.	Examination	Requirement Paragraph(s)
101.	Dimensions and weight not as specified.	3.1, 3.12
102.	Components missing.	3.1
103.	Material not as specified.	3.4
104.	Recovered materials not as specified.	3.4.1
105.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.	3.4.2
106.	Dissimilar metals not in accordance with MIL-STD-889.	3.4.3
107.	Material and material finish or treatment information not available.	3.4.4
108.	Safety provisions missing or not as specified.	3.6
109.	Maintenance provisions not as specified.	3.7
110.	Fasteners not as specified.	3.7
111.	Pneumatic system not as specified.	3.9
112.	Compressor pump not as specified.	3.9.1
113.	Air-intake cleaner missing or not as specified.	3.9.2
114.	Air-discharge manifold not as specified.	3.9.3
115.	Air-delivery hose not as specified.	3.9.5
116.	Nozzles and inflating valves not as specified.	3.9.6
117.	Engine installation not as specified.	3.10
118.	Fuel tank not as specified.	3.10.1
119.	Toolbox missing or not as specified.	3.11
120.	Frame, wheels, or tires not as specified.	3.12
121.	Lubrication not as specified.	3.14
122.	Lubricants not as specified.	3.14.1
123.	Lubrication fittings not as specified.	3.14.2
124.	Pressure-release device not as specified.	3.14.3
125.	Filling, draining, and checking provisions not as specified.	3.14.4
126.	Grease lubrications not as specified.	3.14.5
127.	Oil Lubrication not as specified.	3.14.6
128.	Fungus and moisture resistance not as specified.	3.15

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TABLE I. Examination schedule. (continued)

Defect no.	Examination	Requirement Paragraph(s)
129.	Identification marking missing or not as specified.	3.16
130.	Instruction plates missing or not as specified.	3.17
131.	Treatment and painting not as specified.	3.18
132.	Repair parts or maintenance tools missing.	3.19
133.	Items to be furnished with compressor not as specified.	3.20
134.	Workmanship not as specified.	3.21
135.	Castings and forging not as specified.	3.21.1
136.	Metal fabrication not as specified.	3.21.2
137.	Welding not as specified.	3.21.3
138.	Welders not as specified.	3.21.3.1
139.	Riveted connections not as specified.	3.21.5
140.	Bolted connections not as specified.	3.21.4

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified herein, tests shall be performed with the engine operating at rated speed, and with all components, except the air-delivery hose, nozzles and inflating valves, installed and functioning normally. Prior to tests, the compressor shall be lubricated with oils and greases as specified in 3.14. Oils shall be those designated for use in the ambient temperature at the place of test.

4.6.2.2 Test schedule. Tests shall be in accordance with table II. It is not required that the tests be conducted in the order listed, with the following exceptions:

a. The power consumption, capacity, drop, and air-relief valve tests shall be conducted in the indicated sequence, and shall precede all other tests.
b. The capacity and air-relief valve tests shall be repeated at the conclusion of the endurance test.
c. Running time from the high temperature, low temperature, tilt operation, and fuel tank capacity tests may be included in the accumulated time for the endurance test when operating intervals are as specified in 4.6.2.11.

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TABLE II. Test schedule.

Inspection sequence	First article	Initial production	Quality conformance	Tests	Test Par.	Req. Par.
1	2	3	4	5	6	7
1	X	-	-	Power consumption	4.6.2.3	3.9.1
2	X	X	-	Capacity	4.6.2.4	3.8.1
3	X	-	-	Drop	4.6.2.5	3.8.2
4	X	-	-	Air-relief valve	4.6.2.6	3.9.4
5	X	-	-	High temperature	4.6.2.7	3.5.1
6	X	-	-	Low temperature	4.6.2.8	3.5.1
7	X	-	-	Precipitation	4.6.2.9	3.5.2, 3.9.2
8	X	-	-	Tilt operation	4.6.2.10	3.8.3
9	X	-	-	Endurance	4.6.2.11	3.8.1
10	X	-	-	Fuel tank capacity	4.6.2.12	3.10.1
11	X	-	-	Electromagnetic interference	4.6.2.13	3.13
12	X	-	-	Sound level	4.6.2.14	3.6.1
13	-	-	X	Operational	4.6.2.15	3.8.1, 3.9.4

4.6.2.3 Power consumption.

4.6.2.3.1 Test procedure. Using a dynamometer or a calibrated motor, operate the compressor pump at an input shaft speed of 3600 rpm, with the air-intake cleaner installed and the back pressure on the compressor pump outlet regulated to not less than 6.5 psig. Allow discharge air temperature to stabilize, and then take at least four input power readings at intervals of not less than 10 minutes.

4.6.2.3.2 Failure criteria. An input power requirement in excess of that specified in 3.9.1 shall constitute failure of this test.

4.6.2.4 Capacity. The capacity of the compressor shall be determined in accordance with the ASME Performance Test Code (PTC-9), using either the flow nozzle test method, or the slip test method. The air-relief valve shall be blocked to prevent the passage of air. A minimum of four readings shall be taken at intervals of not less than 10 minutes, after the air-discharge temperature has stabilized. For the slip test, the compressor pump may be driven by an alternate power source. Nonconformance to 3.8.1 shall constitute failure of this test.

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4.6.2.5 Drop. The compressor shall be subjected to a free fall of not less than 18 inches onto a level concrete surface in such a way that the compressor will land initially on its wheels in a level position. Tires shall be inflated to the proper pressure. Evidence of damage, misalignment, or permanent deformation shall constitute failure of this test.

4.6.2.6 Air-relief valve.

4.6.2.6.1 Test procedure. Block the air-discharge ports to prevent passage of air. Install a means to measure manifold pressure. Operate the compressor for no less than 5 minutes and observe full flow operation of the air-relief valve. Then, allow air to flow through the discharge port(s) at such a rate that manifold pressure is maintained at 5.0 psig. Observe air-discharge from the air-relief valve by attaching one end of a hose to the valve discharge and immersing the other end in a vessel of water. Care should be taken to assure that there is no air leakage from other parts of the valve such as adjustment access holes. If airflow through the valve is indicated by bubbles, determine the flow rate by placing the hose into a second vessel filled with water, with its open end down and below the water level in the first vessel. The volume of water displaced by the discharge air from the valve in a period of not less than 10 seconds will determine the discharge rate.

4.6.2.6.2 Failure criteria. Discharge of air through the air-relief valve in excess of 0.25 cfm at a manifold pressure of less than 5 psig, or full flow operation at a manifold pressure greater than 7.5 psig, shall constitute failure of this test.

4.6.2.7 High temperature.

4.6.2.7.1 Test procedure. Subject the compressor to an ambient temperature of not less than 120° F ±5° F until all temperature readings specified herein have stabilized. Start and operate the compressor for 4 hours, while maintaining an ambient temperature of 120° F ±5° F, and with the manifold pressure regulated to not less than 6.5 psig. Record the following data at 30 minute intervals during operation of the compressor:

a. Ambient air temperatures in test cell, 3 to 5 feet from the compressor at three points, 90 degrees apart at crankshaft level.
b. Engine lubricating oil sump temperature.
c. Compressor air-discharge temperature (measured in the discharge air stream, not more than 3 inches from the discharge port).
d. Compressor discharge capacity (capacity may be measured by use of a calibrated flowmeter).
e. Engine rpm.

4.6.2.7.2 Failure criteria. Any damage to the compressor, nonconformance to 3.5.1, or temperature rise greater than that specified in 3.8.1, shall constitute failure of this test.

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4.6.2.8 Low temperature.

4.6.2.8.1 Test procedure. Subject the compressor to an ambient temperature of -25° F ±5° F, until all temperature readings listed in 4.6.2.7.1 have stabilized. Start and operate the compressor for 4 hours, while maintaining an ambient temperature of -25° F ±5° F, and with the manifold pressure regulated to not less than 6.5 psig. Record the data listed in 4.6.2.7.1 at 30 minute intervals during operation of the compressor.

4.6.2.8.2 Failure criteria. Any damage to the compressor, nonconformance to 3.5.1, or temperature rise greater than that specified in 3.8.1, shall constitute failure of this test.

4.6.2.9 Precipitation.

4.6.2.9.1 Test procedure. Subject the compressor to a fresh water spray at a minimum rate of 3 inches per hour directed simultaneously at all four sides and on the top at an angle of approximately 45 degrees from the vertical. The apparatus for producing the spray shall emit water in the form of droplets rather than a fine mist. Expose the compressor to the spray for 1/2 hour while not running and then start and operate the compressor for 1/2 hour at no load while the spray continues.

4.6.2.9.2 Failure criteria. Water leakage into air cleaner or crankcase, or water accumulation on other components which prevents operation shall constitute failure of this test.

4.6.2.10 Tilt operation test procedure. Tilt the compressor 15 degrees from the horizontal about the longitudinal axis and operate in that position for 30 minutes. Repeat this test in each of the four directions (compressor tilted with respect to the longitudinal and transverse axes). Allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel system and then start and operate the compressor for 30 minutes in that position. Any malfunction due to tilted operation such as lack of lubrication, rough or erratic operation, or permanent deformation of any part or component shall constitute failure of this test.

4.6.2.11 Endurance. Operate the compressor for not less than 500 hours continually in periods of not less than 4 hours of running time. For this test, an auxiliary fuel supply system may be used to permit extended operation without refueling. If an auxiliary fuel system is not used, periods of nonoperation of not more than 10 minute duration will be allowed for refueling. The refueling time shall not be counted as operating time, but shall not cause disallowance of the operating period during which it takes place. Oil changes and other preventive maintenance shall be performed during downtime. There shall be no replacement of parts. Adjustments provided by integral devices and repairs which require 1 man-hour or less may be made. The 500 hours of operation shall be accomplished at rated capacity and pressure. Record appropriate remarks, including a record of all stops and adjustments. Record the following data at least once during each 24 hour period. Reading shall be taken only after at least 1 hour of continuous operation:

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a. Air temperature, ambient.
b. Air temperature at discharge.
c. Barometric pressure.
d. Air temperature at compressor intake.
e. Air temperature upstream of nozzle.
f. Air pressure of compressor intake.
g. Nozzle throat diameter.
h. Nozzle coefficient of discharge.
i. Differential pressure across the nozzle.
j. Engine speed (rpm).
k. Compressor unit speed (rpm).
l. Compressor capacity (cfm) computed.

4.6.2.12 Fuel tank capacity. Fill the fuel tank. Start the compressor engine and operate until the engine stops due to lack of fuel. No refueling or auxiliary fuel supply shall be allowed. Insufficient capacity for at least 4 hours of continuous operation shall constitute failure of this test.

4.6.2.13 Electromagnetic interference. The compressor shall be subjected to the tests required by MIL-STD-461 for class III C equipment. Nonconformance to MIL-STD-461 shall constitute failure of this test.

4.6.2.14 Sound level. Place the compressor in a test area which is free of any large reflecting surfaces within 50 feet of the compressor or sound-level measuring equipment. With the compressor operated at rated load, a sound level survey shall be made around the compressor to determine the direction in which the maximum sound levels are propagated so that the test equipment can be properly oriented. The recording microphones shall be positioned in accordance with MIL-STD-1474, category D. The noise level shall be measured with the compressor operated at no load and rated load. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2.14.1 Exterior noise. The compressor shall be tested in accordance with EPA 40 CFR 204 for portable air compressors. Nonconformance to 3.6.1.3 shall constitute failure of this test.

4.6.2.15 Operational. Operate each compressor for not less than 2 hours utilizing all operator controls and determine capacity. Capacity may be measured by use of a calibrated meter. Inability of the compressor to produce 60 cfm, or any malfunction, leakage, or rough or irregular operation shall constitute failure of this test.

4.7 Inspection comparison. The Government may select compressors at any time during the contract production period and subject these compressors to the examination specified in 4.4 and the tests specified in 4.5 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article model and initial production compressors. In addition to any test specified as part of the

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inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compressors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and marking shall be examined and tested in compliance with the quality assurance provisions of MIL-C-3600.

5. PACKAGING

5.1 Preservation, packing, and marking. The air compressors, accessories, repair parts, tools, and publications shall be preserved, packed, and marked in accordance with MIL-C-3600. The level of preservation and the level of packing shall be level A, B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The compressor is intended for use as a portable source of air supply for the inflation of camouflage floats and to maintain the air required to keep pneumatic bridge pontoons fully inflated.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article (see 3.2).
c. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
d. When initial production testing is required and number of compressors to be furnished for initial production testing, when applicable (see 3.3).
e. Color (see 3.18).

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f. Applicable list of repair parts and maintenance tools (see 3.19).
g. When certain spare parts and maintenance tools are required (see 3.19).
h. Degree of preservation and degree of packing required (see 5.1).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.10.

6.4 First article. When a first article inspection is required, the items should be a preproduction compressor. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.5 Lubricants. Selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Unless otherwise authorized by the US Army Belvoir Research and Development Center, lubricants for ground equipment systems shall be restricted to those listed in MIL-HDBK-113.

6.6 Inflating valves. Inflating valves manufactured by Scovill Manufacturing Co., Waterbury, Conn., Schrader Catalogue No. 9927 or equal have been found to conform to the requirements for inflating valves specified in 3.9.6.

6.7 Recovered material. It is encouraged that recovered material be used when practical as long as it meets the requirements of the specification (see 3.4.1).

6.8 Noise limits. If the compressor is designed to meet the noise limit requirement, the weight limit requirement may be exceeded. In view of this, the solicitation process must provide bidders an opportunity to:

a. Address what design aspects (design or shielding) will be required to meet the noise requirement and the cost of each design aspect. This information must be supplied with engineering feasibility data in accordance with MIL-STD-1474.
b. List the calculated weight increase for each noise suppression design aspect. These calculations must be provided for each design aspect.

Before an award is made, a decision on final noise and weight limits must be rendered in conjunction with the Belvoir R&D Center, ATTN: STRBE-NC, Fort Belvoir, Virginia 22060-5606.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - CS

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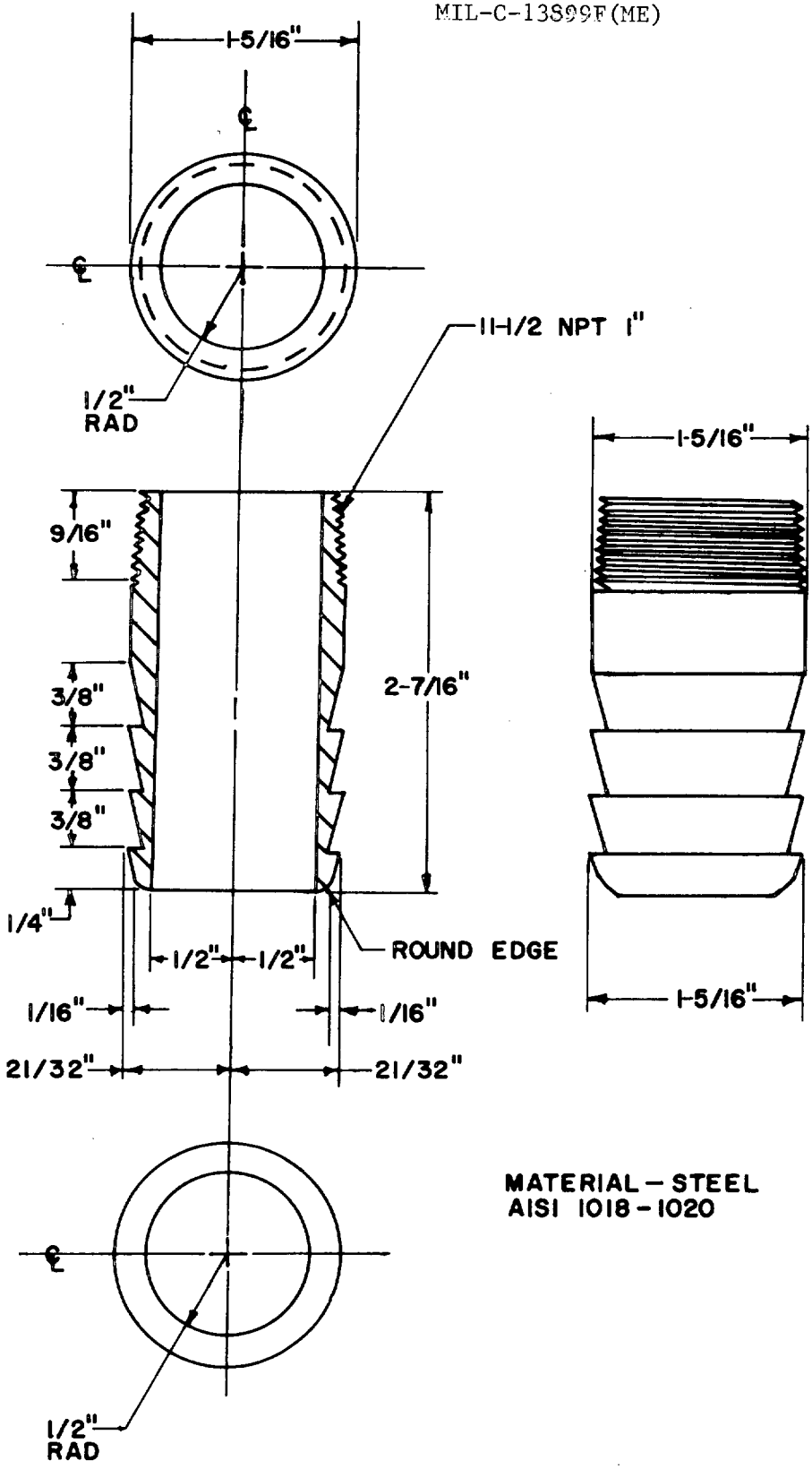


FIGURE 1. –ADAPTER

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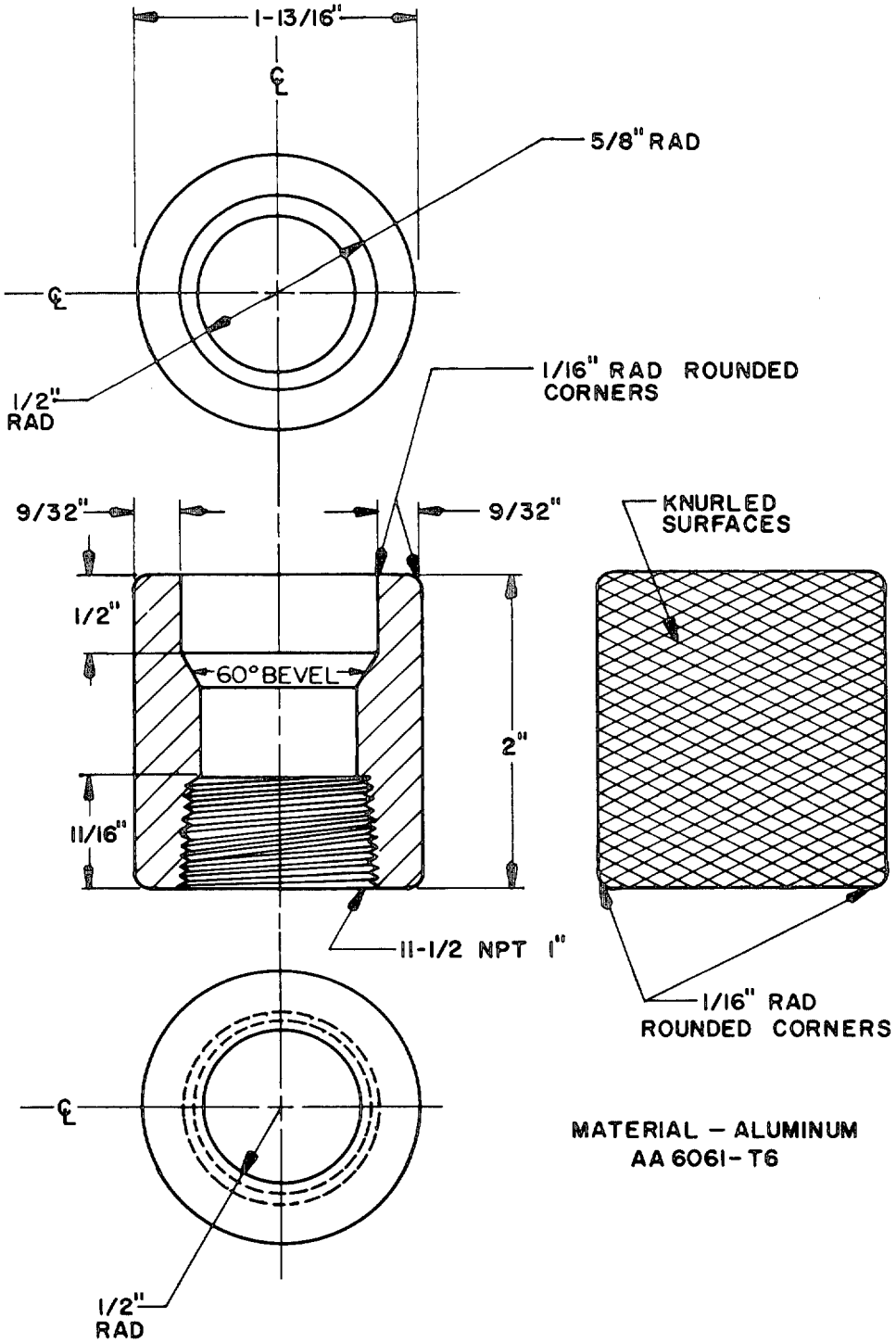


FIGURE 2. – NOZZLE

X-2213A

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